

Wireless-Tag IDO-SMLCD72-V1-2EC - board certification review

Written by kisvegabor on September 09, 2021

IDO-SMLCD72-V1-2EC is powerful board with OpenWRT operating system, 1024x600 7" screen and 1.2 GHz CPU.

🔗 <https://www.youtube.com/embed/ut0-dHNNsD4>

IDO-SMLCD72-V1-2EC is a 7-inch smart display motherboard based on SigmaStar SSD201/SSD202 SoC (ARM Cortex A7 processor), with the main frequency up to 1.2GHz with 256KB L2-Cache. It comes with built-in dual MAC, one PHY, and supporting dual 100MB Ethernet interfaces. It integrates H.264/AVC and H.265/HEVC decoders and supports max. resolution FHD(1920x1080)/60 fps decode.

The board also has audio interface, speaker interface, USB2.0 and WIFI. The manufacturer (Writeless-tag) recommends this board for smart building indoor intercom, smart home central control, elevator floor display, IP network broadcasting device or speech recognition equipment.

IDO-SMLCD72-V1-2EC earned the Professional LVGL board certification which means it's a top tier board with high performance, quality and robustness.



Buy now

The IDO-SMLCD72-V1-2EC board can be purchased from [Alibaba](#).

Specification

CPU and memory

- CPU [Sigmastar SSD202](#) ARM Cortex A7, 1.2 GHz
- RAM 128MB
- Flash 128MB
- GPU None
- Operating system [OpenWrt](#)

Display

- Resolution 1024*600
- Display size 7"
- Color depth 24bit,RGB888

- **Technology** IPS
- **DPI** 170 px/inch
- **Touch pad** Capacitive

Connectivity

- Wifi: Wireless card onboard + Extra ESP32-WROOM-32U for Alot
- Bluetooth: Extra ESP32-WROOM-32U for Alot
- 4 GPIO available
- 100Mbps Ethernet
- 4pin speaker port
- Analog microphone
- Digital microphone
- TF card
- USB 2.0
- 4 UART port (uart0 for debug and uart3 for the esp32)
- 4G cat1 module
- RTC onboard
- 1080P video decode or stream media can be played via LVGL

Others

- **Power supply** USB-C (5V)
-

Performance

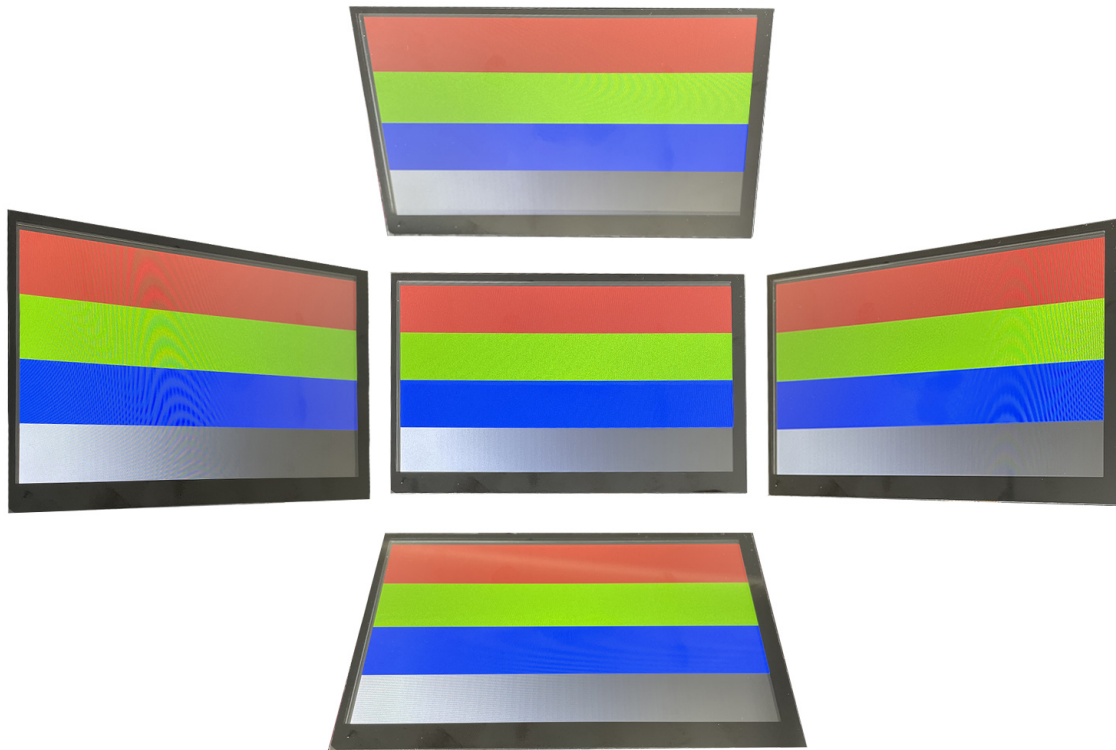
IDO-SMLCD72-V1-2EC is a powerful computer with OpenWrt (Linux). Even without any GPU, using only the pure 1.2 GHz clock of the CPU the FPS has never dropped below 60 during the benchmark. The driver uses the the standard `dev/fb` to simply copy the LVGL rendered image to the framebuffer.

The booting time is very good. The demo started in 8 seconds from power up.

Quality

Display

This board has a large beautiful display with great pixel density.



Touchpad

IDO-SMLCD72-V1-2EC comes with a capacitive touchpad. Therefore it recognizes touches with a good precision and provides a smartphone like experience. The drawback is that the touchpad can not be used in gloves or with a pen.

Robustness

IDO-SMLCD72-V1-2EC is designed to be placed into an end product. It's massive and has many holes to mount the board.

Development

On the products of Wireless-Tag you can develop 2 different ways:

1. Write code and built it. See [here](#)
2. Use their custom UI Editor tool called [8ms](#).

Build and deploy

A getting started project can be found on [GitHub](#). The development instruction in the README are in Chinese so we summarize them here. (Mainly the translation of the README)

Installation dependencies

(Verified on Ubuntu 16.04.7 64-bit)

```
sudo apt-get install subversion build-essential libncurses5-dev zlib1g-dev gawk git ccache \
    gettext libssl-dev xsltproc libxml-parser-perl \
    gengetopt default-jre-headless ocaml-nox sharutils textinfo
sudo dpkg --add-architecture i386
sudo apt-get update
sudo apt-get install zlib1g:i386 libstdc++6:i386 libc6:i386 libc6-dev-i386
```

Install toolchain

Download the toolchain from [here](#).

- Unzip the toolchain

```
sudo tar wt-gcc-arm-8.2-2018.08-x86_64-arm-linux-gnueabi.tar.gz -xvf -C /opt/
```

- Set environment variables: modify `~/.profile` file, add the following line to the end of the file

```
PATH="/opt/gcc-arm-8.2-2018.08-x86_64-arm-linux-gnueabi/bin:$PATH"
```

- Update the environment variables
`source ~/.profile`
- Test cross toolchain
`arm-linux-gnueabi-gcc --version`

Get and build the project

```
git clone https://github.com/wireless-tag-com/8ms-sstar.git
cd 8ms-sstar
make -j
```

Deploy

You can use either the serial port or the Ethernet port to connect to the board.

We have used the Ethernet port via `telnet` :

```
telnet 192.168.1.1
```

The password is `admin` .

After you can use all the normal Linux commands. To copy the built project to the board we have used `scp` . On the development PC use the following command to copy the created binary to the boards root folder.

```
scp ./bin/demo root@192.168.1.1:/
```

Going back to the `telnet` session and issuing `./demo` will start the demo.

If you place the `demo` to `usr/sbin` the demo will start automatically on start up.

Editor

Wireless-Tag's online editor is available at 8ms.xyz. It's drag and drop editor allowing you to create UIs without writing a single line of code. Besides easily putting together a UI you can use Blockly to implement the business logic.

After signing up and logging in click the red "Workbench" button on to right corner.



After click the "New" button on the right, Select "Sigmaster" and "SSD201+SPI NAND + CC0702I50R(1024*600)" hardware.

New Project Wizard

New

Sigmaster

Smart HD Display Controller

ESP32

IOT WIFI+蓝牙+彩屏

* Project Name:

* Hardware:

SSD201+SPI NAND + CC0702I50R(1024*600) ▼

Theme:

default ▼

Description:

Tags:

Click confirm and you are ready play with the UI.

In the "Layout" menu you can select "Blockly" to add some logic too.

From the "Compile" menu you can download the binary, the source code or start an online preview.

Conclusion

IDO-SMLCD72-V1-2EC is an amazing board. Using OpenWRT (Linux) the development, development and maintenance are very simple. Besides the the performance and the robustness of the board are also great.

Due to 8ms editor you don't need to be a professional developer to get started with this product.



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